YUVRAJ DHANRAJ KALE

Full Stack Developer | Software Development Engineer | ML Engineer

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PROFESSIONAL SUMMARY

Results-driven B.Tech Information Technology student at NMIMS University (CGPA: 8.2) with strong foundations in full-stack engineering, machine learning, and scalable system design. Proficient in building AI-driven platforms, cybersecurity tools, and production-grade web applications using Java Spring Boot, Python, React.js, SQL, and MongoDB. Experienced in designing RESTful APIs, predictive analytics systems, and responsive frontend architectures aligned with FAANG-level software engineering standards. Seeking high-impact internship roles in Software Development Engineering, Full Stack Development, or Machine Learning Engineering.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, C, C++, HTML5, CSS3, SAS

Frontend: React.js, Responsive UI Design, Component-Based Architecture, DOM Manipulation, REST API Integration

Backend: Spring Boot, Node.js, Express.js, FastAPI, RESTful Architecture, Authentication Systems

Databases: SQL, MongoDB, DBMS, Query Optimization, Data Warehousing, ETL Concepts, OLAP/OLTP

AI / ML: Scikit-learn, Pandas, NumPy, TensorFlow (Foundational), Matplotlib, Feature Engineering, Predictive Analytics, Classification Models

Tools & DevOps: Git, GitHub, Postman, VS Code, Linux Basics, Vercel, Netlify, GitHub Pages, Basic Cloud Deployment

Core CS: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, Software Engineering, System Design

Learning Next: Docker, Kubernetes, AWS, Redis, Kafka, Microservices Architecture, CI/CD Pipelines, Distributed Systems

EXPERIENCE

Ajay Infotech | Web Development Instructor *Pune, Maharashtra | 2 Months*

- Mentored 20+ students in frontend web development technologies including HTML5, CSS3, JavaScript, and React.js, driving measurable improvement in hands-on project completion rates.
- Designed and delivered structured coding modules focused on responsive UI development, debugging techniques, component-based architecture, and REST API integration, increasing practical implementation efficiency.
- Guided students in building and deploying mini-projects using Git and GitHub workflows, strengthening collaborative version control practices and real-world software development readiness.
- Introduced project-based learning methodologies and real-world case studies, improving student engagement and coding confidence across frontend optimization and DOM manipulation concepts.
- Coached learners in understanding modern React patterns including reusable component design, state management, and frontend deployment workflows using Vercel and Netlify.

PROJECTS

Digital Forensics & Incident Response Platform | *Python, React.js, MongoDB, REST APIs, Log Analysis*

- Engineered a full-stack digital forensics and incident response platform capable of monitoring suspicious system activities, analyzing security event logs, and visualizing threat intelligence data in real time.
- Developed automated evidence collection workflows using Python-based log parsing pipelines, reducing manual forensic investigation overhead and improving analysis turnaround efficiency by an estimated 40%.
- Implemented anomaly detection mechanisms leveraging rule-based log analysis algorithms to identify irregular access patterns, unauthorized executions, and potential intrusion indicators.
- Designed an interactive React.js dashboard for visualizing incident reports, forensic event timelines, and threat activity metrics, enabling faster security incident triage and response decision-making.
- Integrated structured forensic evidence storage using MongoDB, implementing schema-based data organization for efficient querying, retrieval, and audit trail management.

AI-Based Risk Assessment & Fraud Detection System | *Python, Scikit-learn, Pandas, NumPy, Machine Learning*

- Built a supervised machine learning risk assessment model for predictive fraud detection and financial risk scoring, achieving classification accuracy improvements through iterative feature engineering and model tuning.
- Processed and cleaned large structured datasets using Pandas and NumPy, implementing normalization, outlier removal, missing-value imputation, and categorical encoding to improve model training quality.
- Implemented and benchmarked multiple classification algorithms including Random Forest, Logistic Regression, and Gradient Boosting, selecting the optimal model based on precision, recall, and F1-score evaluation.
- Optimized model performance through hyperparameter tuning, cross-validation strategies, and confusion matrix analysis, enabling reliable risk-scoring outputs aligned with fintech enterprise requirements.

- Designed scalable preprocessing and inference pipelines suitable for future integration into real-time fraud monitoring and enterprise analytics systems.

Spotify-Inspired Music Streaming Web Application | *React.js, JavaScript, HTML5, CSS3*

- Developed a responsive, production-quality Spotify-inspired music streaming frontend implementing modern UI/UX design principles, dynamic playlist rendering, and cross-device compatibility.
- Architected a modular React.js component library including reusable player controls, album grid views, and sidebar navigation, improving codebase maintainability and future feature extensibility.
- Implemented dynamic media playback, playlist state management, and interactive UI controls using React hooks and JavaScript event-driven architecture, delivering seamless user interaction workflows.
- Optimized frontend rendering performance through component memoization and efficient state updates, reducing unnecessary re-renders and improving application responsiveness.
- Deployed the application using modern hosting workflows (Vercel/GitHub Pages) and maintained clean version control practices using Git branching strategies.

Machine Learning Classification & Prediction Models | *Python, TensorFlow, Scikit-learn, Pandas, Matplotlib*

- Developed multiple supervised machine learning models for classification and regression tasks across diverse datasets, implementing end-to-end pipelines from raw data ingestion to model deployment readiness.
- Built comprehensive preprocessing pipelines including feature normalization, dimensionality reduction, train-test splitting, and class imbalance handling using SMOTE and resampling techniques.
- Evaluated model performance using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis to ensure reliable generalization across validation datasets.
- Applied data visualization techniques using Matplotlib and Seaborn to analyze feature distributions, training convergence trends, and model prediction behavior for interpretability insights.

Distributed Microservices E-Commerce Platform [In Progress] | *Java Spring Boot, React.js, MongoDB, Docker, Redis*

- Designing a scalable microservices-based e-commerce platform with independent services for product management, authentication, order processing, and payment integration using Spring Boot and RESTful API architecture.
- Planning Redis-based caching layers and API gateway routing for optimized service-to-service communication, reduced database load, and sub-100ms response latency targets.
- Preparing containerized deployment workflows using Docker for improved service portability, environment consistency, and infrastructure scalability across development and production environments.

AI-Powered Resume Screening System [Planned] | *Python, NLP, FastAPI, React.js, MongoDB*

- Designing an ATS-compatible resume screening platform leveraging NLP pipelines for automated semantic skill extraction, keyword matching, and intelligent candidate ranking aligned with job description requirements.
- Planning a FastAPI-based backend with asynchronous processing, MongoDB storage, and a React.js recruiter dashboard for real-time candidate scoring visualization and hiring insights.

Real-Time Cybersecurity Threat Detection Dashboard [Planned] | *Python, Apache Kafka, ELK Stack, React.js*

- Conceptualizing a real-time security analytics platform integrating Kafka-based streaming pipelines for high-volume log ingestion, anomaly detection, and automated threat alert generation.
- Planning ELK Stack (Elasticsearch, Logstash, Kibana) integration for scalable log indexing, full-text search, and operational security analytics dashboards suitable for enterprise-level SOC environments.

EDUCATION

NMIMS University *Pune, Maharashtra | 2024 – 2028*

Bachelor of Technology (B.Tech) — Information Technology | CGPA: 8.2/10.0

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Machine Learning, Data Warehouse & Mining, Computer Networks, Software Engineering

CERTIFICATIONS

- NPTEL — C++ for Modern Computing (National Programme on Technology Enhanced Learning, IIT-backed certification)
- Machine Learning Model Development — Academic & Self-Learning Projects (Scikit-learn, TensorFlow, Pandas pipelines)
- Frontend Development & React.js Application Engineering — Verified through production project deployments

ACHIEVEMENTS & LEADERSHIP

- Selected and participated in Smart India Hackathon (SIH), India's largest national-level hackathon, delivering a working prototype under a 36-hour engineering sprint focused on AI and software engineering solutions.
- Competed in multiple college-level and inter-institutional hackathons focused on AI, full-stack development, and cybersecurity problem-solving, demonstrating strong analytical and rapid prototyping capabilities.
- Built and deployed 5+ end-to-end technical projects spanning machine learning, full-stack web engineering, and cybersecurity forensic analysis — demonstrating consistent initiative beyond academic requirements.

- Maintained a strong CGPA of 8.2 while concurrently undertaking real-world project development, technical mentorship responsibilities, and competitive programming preparation.
- Mentored 20+ junior developers at Ajay Infotech, strengthening collaborative engineering culture and building measurable improvements in student coding proficiency and project delivery outcomes.